GENERAL SCIENCE

TREND ANALYSIS (2016-2014)

PHYSICS

S.No.	Chapter Name	2016 (I)	2015 (II)	2015 (I)	2014 (II)
1	Mechanics	2	4	6	2
2	Heat	1	1	-	-
3	Properties of Matter	-	-	-	-
4	Waves	3	-	-	-
5	Optics	1	2	1	3
6	Electricity	1	1	3	2
7	Modern Physics	-	1	1	-
	Total	**8**	**9**	**11**	**7**

CHEMISTRY

S.No.	Chapter Name	2016 (I)	2015 (II)	2015 (I)	2014 (II)
1	General Chemistry	-	-	1	-
2	Atomic Structure	1	2	1	3
3	Radioactivity	1	-	-	1
4	Chemical Binding and Redox Reactions	-	-	-	-
5	Equilibrium , Gas Law and Catalyst	1	-	-	-
6	Solution	-	-	-	1
7	Acid, Base and Salt	1	-	-	1
8	Thermodynamics and Colloids	2	-	1	-
9	Electrochemistry	-	-	2	-
10	Inorganic Chemistry	3	3	3	1
11	Organic Chemistry	1	-	-	4
12	Man Made Materials	1	1	1	1
13	Environment and Its Pollution	-	3	2	1
	Total	**11**	**9**	**11**	**13**

BIOLOGY

S.No.	Chapter Name	2016 (I)	2015 (II)	2015 (I)	2014 (II)
1	Introduction	-	-	-	-
2	Cell and Inheritance	1	-	1	1
3	Origin of Life and Its Growth	-	2	1	1
4	Classifications of Plants and Animals	-	-	2	3
5	The Plant Life	1	1	1	1
6	Animal Physiology	8	2	3	3
7	Ecology and Environment	-	1	2	-
8	Applied Biology	1	4	-	1
	Total	**11**	**10**	**10**	**10**

01

PHYSICS

After analysing the previous year question papers, we have seen that 40-45 questions are asked from science section, out of these 20-22 questions are asked from Physics. From physics section, around 2 to 3 questions each are asked from topics like motion, sound, electric current and 1 to 2 questions each are asked from topics like optics, modern physics and heat.

MEASUREMENT, MOTION, WORK, ENERGY AND POWER

PHYSICAL QUANTITIES

All the quantities which can be measured directly or indirectly in terms of which law of physics are described and whose measurement is necessary, are called physical quantities. e.g. velocity, time, mass, etc.

Unit

The standard amount of a physical quantity chosen to measure the physical quantity of same kind is called a physical unit.

Fundamental and Derived Units

The unit of a defined set of physical quantities called fundamental quantities are known as fundamental units and the units for all other physical quantities, except fundamental quantities are known as derived units.

e.g. unit of speed can be derived from fundamental units as,

$$\text{Unit of speed} = \frac{\text{Unit of distance}}{\text{Unit of time}} = \frac{\text{m}}{\text{s}} = \text{ms}^{-1}$$

SI System

It is based on the following seven basic units and two supplementary units.

Name of Quantities	*Name of Units*	*Name of Quantities*	*Name of Units*
Basic Units			
Length	Metre	Thermodynamic temperature	Kelvin
Mass	Kilogram	Luminous intensity	Candela
Time	Second	Quantity of matter	Mole
Electric current	Ampere		
Supplementary Units			
Plane angle	Radian	Solid angle	Steradian

SCALAR AND VECTOR QUANTITIES

- Physical quantities which have magnitude only and no direction are called scalar quantities.
  e.g. mass, speed, volume, work, time, power, energy, etc.
- Physical quantities which have magnitude and direction both and which obey triangle law are called vector quantities.
  e.g. displacement, velocity, acceleration, force, momentum, torque, etc.

MECHANICS

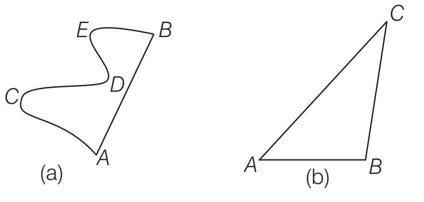
Mechanics deals with the study of motion of particles, rigid and deformable bodies and general system of particles.

Motion

A body is said to be in motion when its position changes continuously with respect to a stationary body taken as a reference point.

Distance and Displacement

- The **distance** covered by a body is the actual length of the path travelled by the body between the initial position and final position.
  - It is always positive.
  - It is a scalar quantity which has magnitude only. Its unit is metre.
- The **displacement** of a particle is the change in position of the particle in a particular direction and is given by a vector drawn from its initial position to its final position.
  - Displacement may be positive, negative or zero.
  - It is a vector quantity and its unit is also metre.
  - The magnitude of displacement may or may not be equal to the path length travelled by an object.
  - Displacement ≤ Distance



  - In Fig. (a) a particle starts from A and reach B through points C, D and E, then AB is the displacement and $(AC + CD + DE + EB)$ is the distance travelled.
  - Also, in Fig. (b), a particle starts from A and reach to C through point B, then AC is displacement and distance covered is $(AB + BC)$.

Speed

The path length or the distance covered by an object divided by the time taken by the object to cover that distance is called the speed of that object. Its unit is m/s or km/h.

$$\text{Speed} = \frac{\text{Distance travelled}}{\text{Time taken}}$$

- An object is said to be moving with a **uniform speed**, if it covers equal distances in equal intervals of time.
- An object is said to be moving with a **variable speed**, if it covers equal distances in unequal intervals of time or unequal distances in equal intervals of time.

Average Speed

- If a motion is the ratio of the total distance travelled by the object to the total time taken.
  i.e. average speed, $v_{\text{av}} = \dfrac{s_1 + s_2 + s_3 + \dots}{t_1 + t_2 + t_3 + \dots}$
- If the body covers equal distances with different speeds, then $v_{\text{av}} = \dfrac{2v_1 v_2}{v_1 + v_2}$

  When the speed of an object is constantly changing, then the speed at a particular moment or instant of time is called the **instantaneous speed** of that object.

Velocity

The rate of change in position or displacement of a body with time is called the velocity of that body. It is a vector quantity.

$$\text{Velocity} = \frac{\text{Displacement in particular direction}}{\text{Time taken}}$$

- A body is said to be moving with **uniform velocity**, if equal displacements of the body take place in same direction in equal intervals of time.
- If a body moves in such a way that its speed or the direction or both changes with time, the body is said to have **variable velocity**.

Average Velocity

- If a body is defined as the net displacement divided by the time taken for displacement.

$$v_{av} = \frac{x_2 - x_1}{t_2 - t_1}$$

  As net displacement $= x_2 - x_1$ and time taken $= t_2 - t_1$
- Average velocity could be zero, positive or negative but average speed is always positive for moving body. When a particle returns to the starting point, its average velocity is zero but average speed is not zero.

Relative Velocity

- When two bodies are moving in the straight line, the speed (or velocity) of one with respect to another is known as its relative speed (or velocity).
- When two bodies are moving in the same direction, then relative velocity $= v_1 - v_2$. When the two bodies are moving in opposite directions, then relative velocity $= v_1 + v_2$.

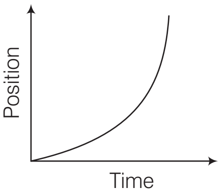
Acceleration

- It is the rate of change of velocity with respect to time. Its SI unit is m/s^2 and CGS is cm/s^2. It is a vector quantity.
- An object is said to be moving with **uniform acceleration**, if its velocity changes by equal amounts in equal intervals of time.
- An object is said to be moving with a **variable acceleration**, if its velocity changes by unequal amounts in equal intervals of time.
- If the velocity of an object increases without change in direction, it is said to be moving with positive acceleration.
- If the velocity of an object decreases without change in direction, the object is said to be moving with negative acceleration or deceleration or retardation.
- Acceleration of an object is zero, if it is at rest or moving with uniform velocity.

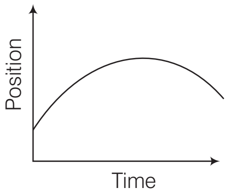
Graphical Representation of Motion

1. Position–Time Graphs

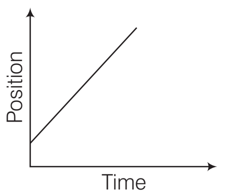
(*i*) When distance covered by a moving object goes on increasing with time, the object is said to have positive acceleration (i.e. velocity increases with time).



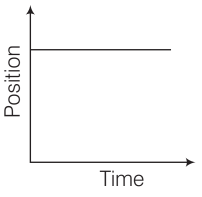
(*ii*) When distance covered by a moving object goes on decreasing with time, the object is said to have negative acceleration (i.e. velocity decreases with time).



(*iii*) When the moving object covers equal distance in equal time, the object is said to have zero acceleration i.e. uniform velocity.

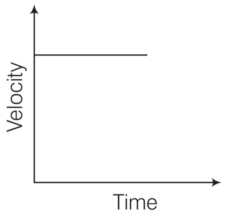


(*iv*) When the position of object does not change with time, the object is at rest (i.e. velocity is zero).

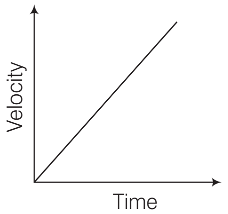


2. Velocity-Time Graphs

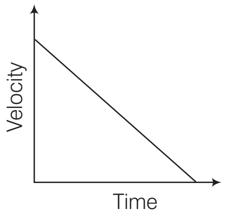
(*i*) In case of zero acceleration, the velocity of the object does not change with time.



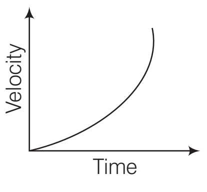
(*ii*) If the object is moving with uniform positive acceleration having zero initial velocity, then the velocity-time graph is a straight line starting from origin.



(*iii*) In case of negative and constant acceleration, the velocity of the object decreases linearly with time.



(*iv*) When the acceleration of the object is increasing, line bending towards velocity axis represent the increasing acceleration in the body.



(*v*) When the acceleration of the object is decreasing, line bending towards time axis represents the decreasing acceleration in the body.